

DSC 40B Discussion 1 Worksheet Version A

Friday, April 3, 2026

Imagine you are a tutor, and you need to explain something to someone who has only a vague understanding of what's going on. Do your best to get the point across. **ALTERNATE**.

If you're not sure, ask your peer if they know how to explain it, then REPEAT the explanation back. Remember, the point of the exercise is to **vocalize** your thoughts.

If you are both unsure, make sure to come to/tutor for help! Do not use ChatGPT or any other LLM! The point of discussion is to prepare you for the exams.

Have one person in the pair open the Discussion 1 Gradescope assignment. After both you **and** your partner finish a problem **and explain it**, answer the corresponding multiple choice/short answer questions for both versions.. You will receive instant feedback on the questions, whether you are correct or not. If your answer is incorrect, please go back to the question to understand it and feel free to ask tutors about anything you are confused about. Once you have finished the entire worksheet or discussion is ending, turn in your assignment and add your partner to the submission.

Quick Icebreaker

Introduce yourself to your partner by sharing name, year, major, etc. Ask each other what you did over spring break. (Do not spend more than 2 minutes!)

Questions

1. Given a list of unique integers, find the **second smallest** number.
 - 1.1 Describe the brute force algorithm. Explain why the brute force algorithm **is** indeed a brute force.
 - 1.2 What is its runtime?
 - 1.3 Come up with an optimal solution for the same problem.
 - 1.4 What is its runtime?

MOVE TO THE SIMILAR QUESTION IN VERSION B.

2. Give one disadvantage of timing programs to determine algorithm efficiency that is different from your partner. Explain in 2-3 sentences why it is a disadvantage.

MOVE TO QUESTION 3 FOR VERSION A.

3. Using the code below:

- 3.1 Fill in the table.

Code	Runtime per line	# of executions
<pre>def count_positives(nums): count = 0 for x in nums: if x > 0: count += 1 return count</pre>	N/A	N/A

3.2 Solve for the runtime $T(n)$ of `count_positives`. Explain your work.

MOVE TO THE SIMILAR QUESTION FOR VERSION B.

4. What is the runtime of the code below. State your answer using Θ notation. *Make sure to explain the reason for your answer.*

```
def runtime_1(n):
    count = 0
    while n > 0:
        n = n // 2
        count += 1
    return count
```

MOVE TO QUESTION 5 FOR VERSION A.

5. Given the code below, the runtime table has been filled out. However, there are **errors** in the table. Identify and fix the errors in the table. Once you have fixed the errors, calculate the runtime of `some_kind_of_counting` and express it in Θ notation.

Code	Runtime per line	# of executions	Line #
<pre>def some_kind_of_counting(n): count = 0 while n > 0: for i in range(n): count += 1</pre>	N/A	N/A	
	c_1	1	1
	c_2	$n + 1$	2
	c_3	$n^2 + 1$	3
	c_4	n^2	4

<pre> n = n - 1 return count </pre>	n	n	5
	c_5	1	6

MOVE TO THE SIMILAR QUESTION IN VERSION B.

6. What is the complexity for the following expressions in terms of Θ ? Show and explain your work.

$$1 + 2 + 3 + 4 + 5 + \dots + 16n = ?$$

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Quick Icebreaker

Introduce yourself to your partner by sharing name, year, major, etc. Ask each other what you did over spring break. (Do not spend more than 2 minutes!)

Questions

1. Given a list of integers, determine whether the list contains **duplicate integers**.
 - 1.1 Describe the brute force algorithm. Explain why the brute force algorithm **is** indeed a brute force.
 - 1.2 What is its runtime?
 - 1.3 Come up with an optimal solution for the same problem.
 - 1.4 What is its runtime?

MOVE TO QUESTION 2 IN VERSION B.

2. Give one disadvantage of timing programs to determine algorithm efficiency that is different from your partner. Explain in 2-3 sentences why it is a disadvantage.

MOVE TO THE SIMILAR QUESTION IN VERSION A.

3. Using the code below:

3.1 Fill in the table.

Code	Runtime per line	# of executions
<pre>def count_odds(nums): count = 0 for x in nums: if x % 2 == 1: count += 1 return count</pre>	N/A	N/A

3.2 Solve for the runtime $T(n)$ of `count_odds`. Explain your work.

MOVE TO QUESTION 4 IN VERSION B.

4. What is the runtime of the code below. State your answer using Θ notation.

```
def runtime_2(n):  
    counter = 1  
    while counter < n:  
        for i in range(n):  
            print('I love puppies!')  
        counter = counter * 2  
    return "So cute!"
```

MOVE TO THE SIMILAR QUESTION IN VERSION A.

5. Given the code below, the runtime table has been filled out. However, there are errors in the table. Identify and fix the errors in the table. Once you have fixed the errors, calculate the runtime of `some_kind_of_sum` and express it in Θ notation.

Code	Runtime per line	# of executions	Line #
<pre>def some_kind_of_sum(lst):</pre>	N/A	N/A	

<pre> count = 0 for i in range(len(lst)): for j in range(i): count += sum(lst) if lst[i] == 0: print('Zero!') return count </pre>	c_1	1	1
	c_2	$n + 1$	2
	c_3	$(n + 1)^2$	3
	c_4	$n^2 + n$	4
	c_5	$n^2 + n$	5
	c_6	$n^2 + n$	6
	c_7	1	7

MOVE TO QUESTION 6 IN VERSION B.

6. What is the complexity for the following expressions in terms of Θ ? Show and explain your work.

$$\log n + \log n^2 + \log n^3 + \log n^4 + \dots + \log n^n = ?$$